THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

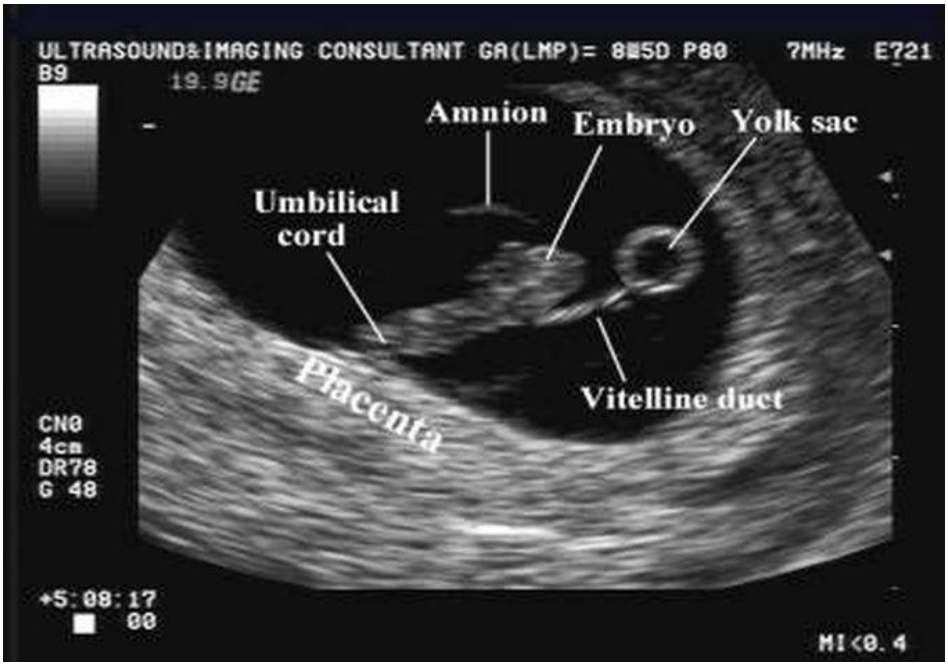
1st Trimester Protocol

Note that the first trimester has three stages protocol. *Images don't belong to CURE; they are taken from online.*

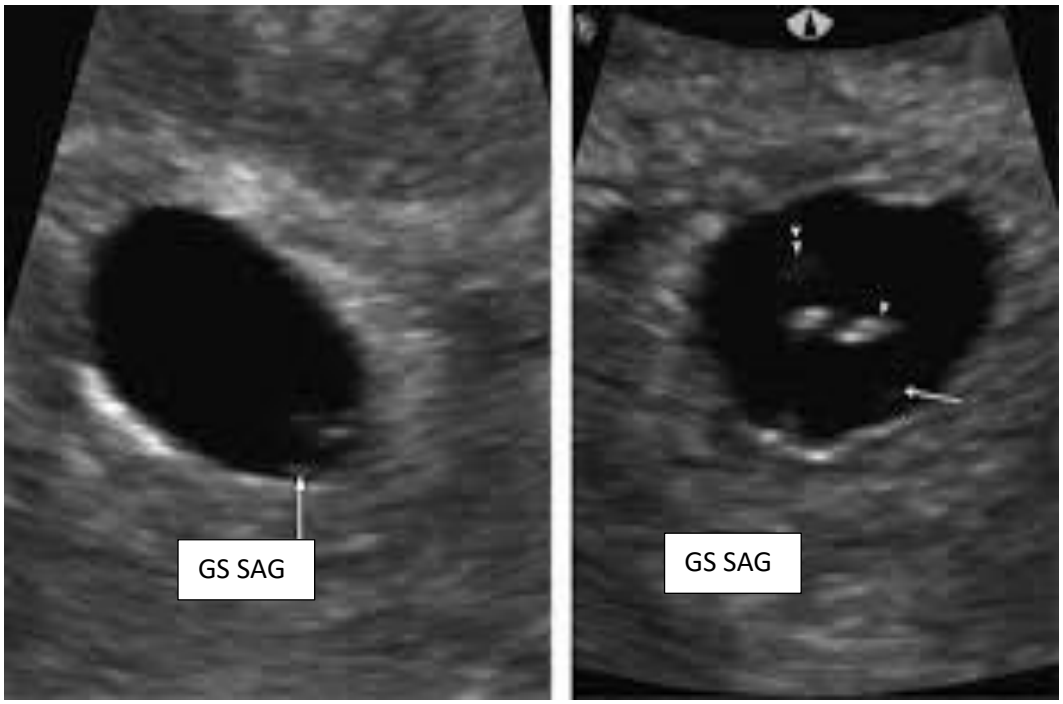
A) Early Stage: 5-8 weeks gestation

You will still need to do the full protocol for the female pelvic ultrasound (Uterus/Ovaries) first before you do this. The following can be taken from an abdominal and/or transvaginal ultrasound approaches, it depends on the facility you work for):

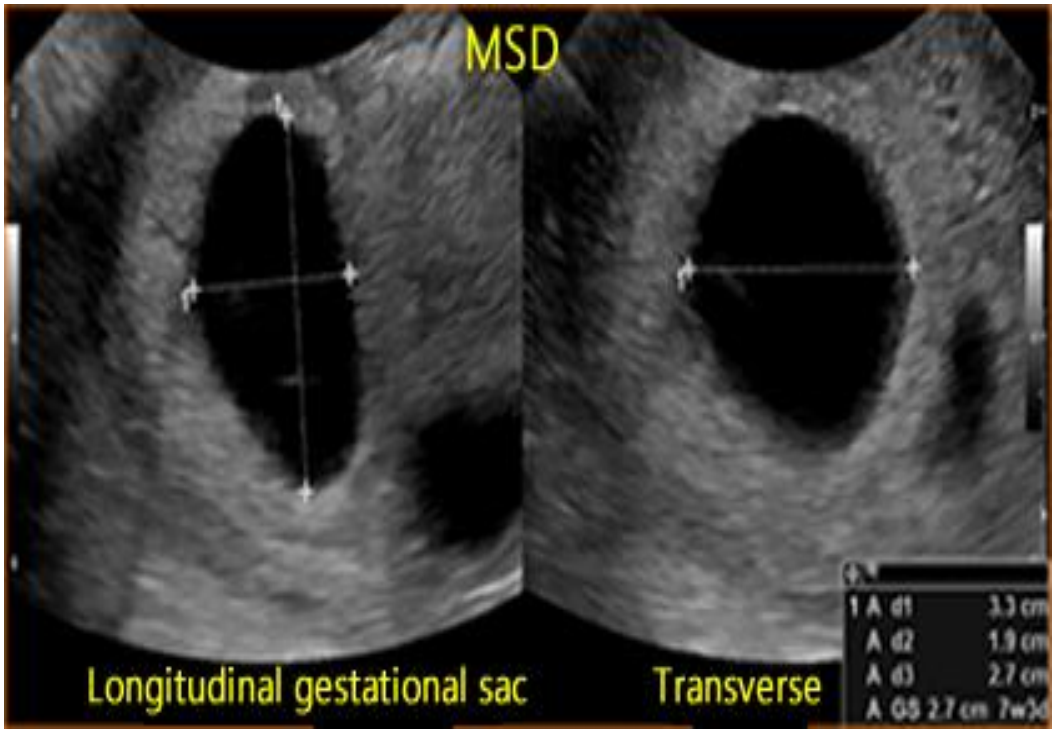
You will probably see this image when you first start scanning



1. **Take a picture in dual in SAG and TRV of the Gestational Sac w/o Measurements**



2. **Take a picture in dual in SAG and TRV of the Gestational Sac w/ Measurement**

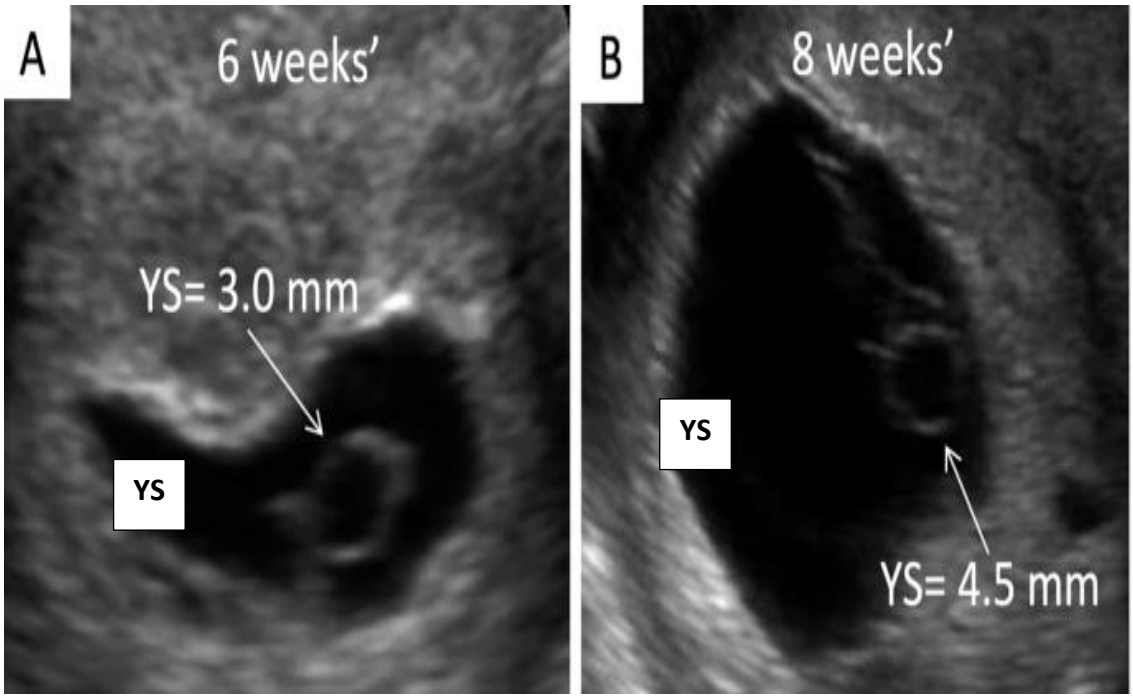


Note:

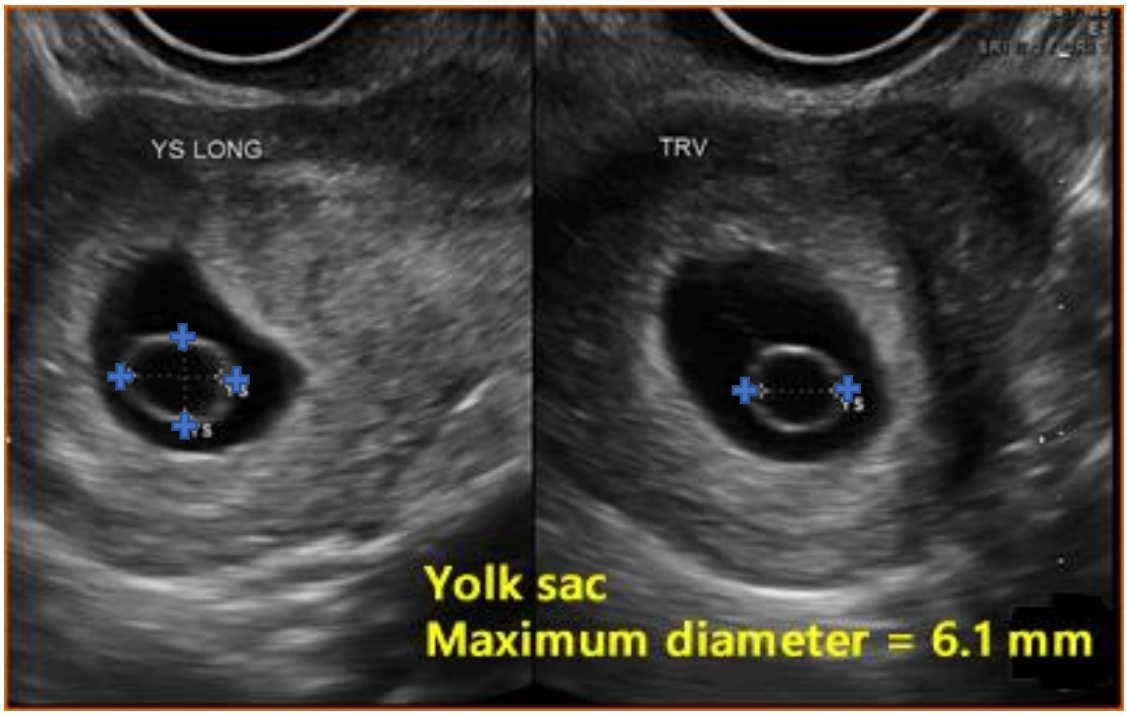
MSD-

Mean sac diameter

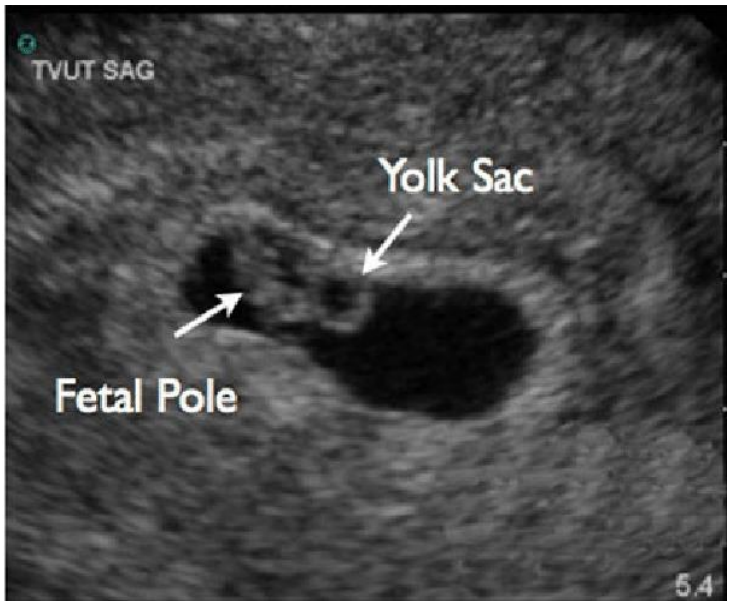
3. Take a picture of the Yok Sac in SAG and TRV w/o Measurement

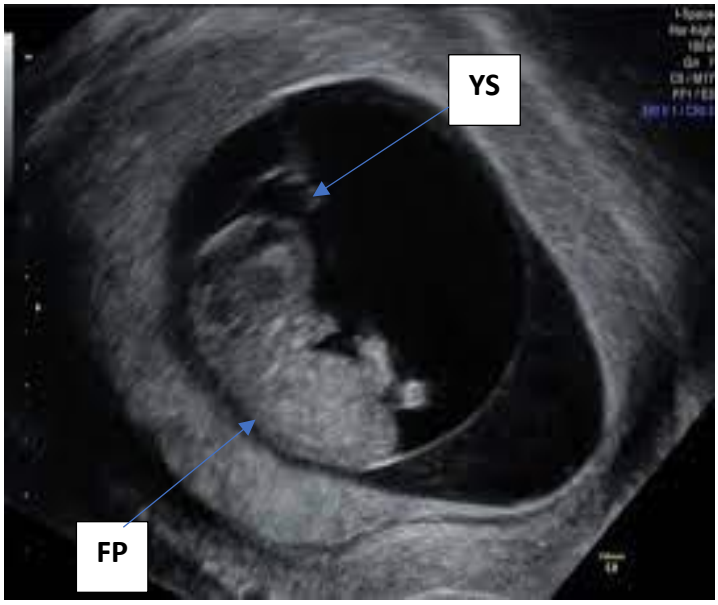


4. Take a picture of the Yok Sac in SAG and TRV of w/ Measurement (outer to outer)



5. Take a picture of the Embryo (FP-Fetal Pole) w/o Measurement. Note the embryo can differ in size





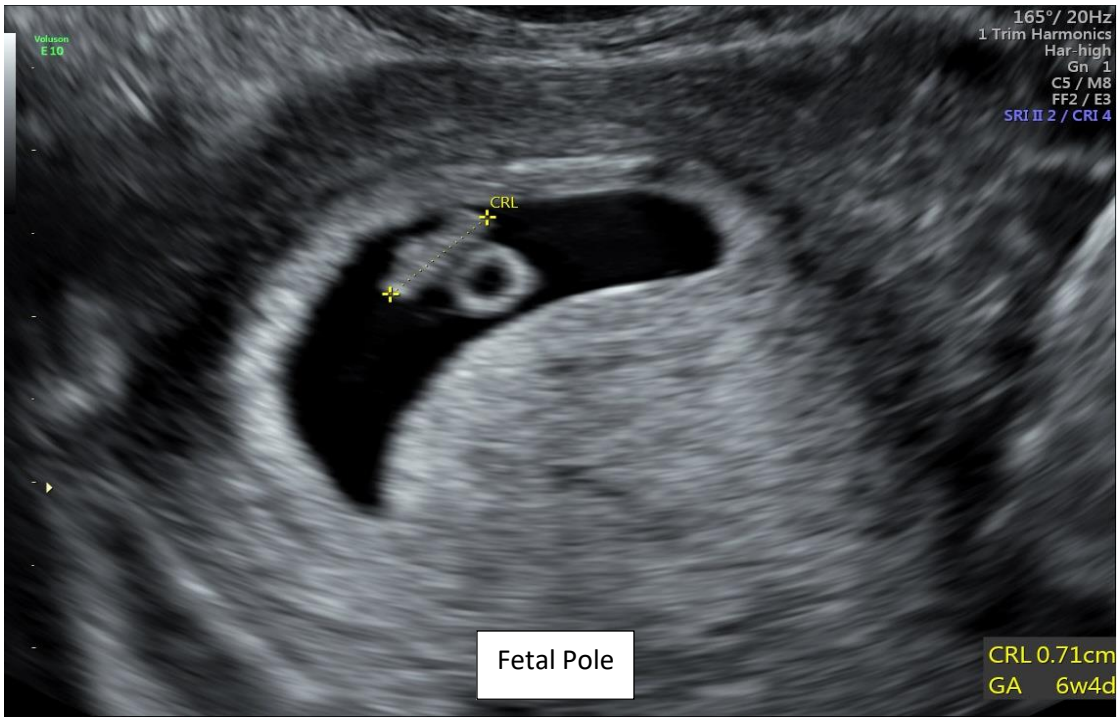
6. Take a picture of the Embryo/Fetal Pole w/ Measurement (it does not matter if it is in SAG or TRV, get the longest length).

Note:

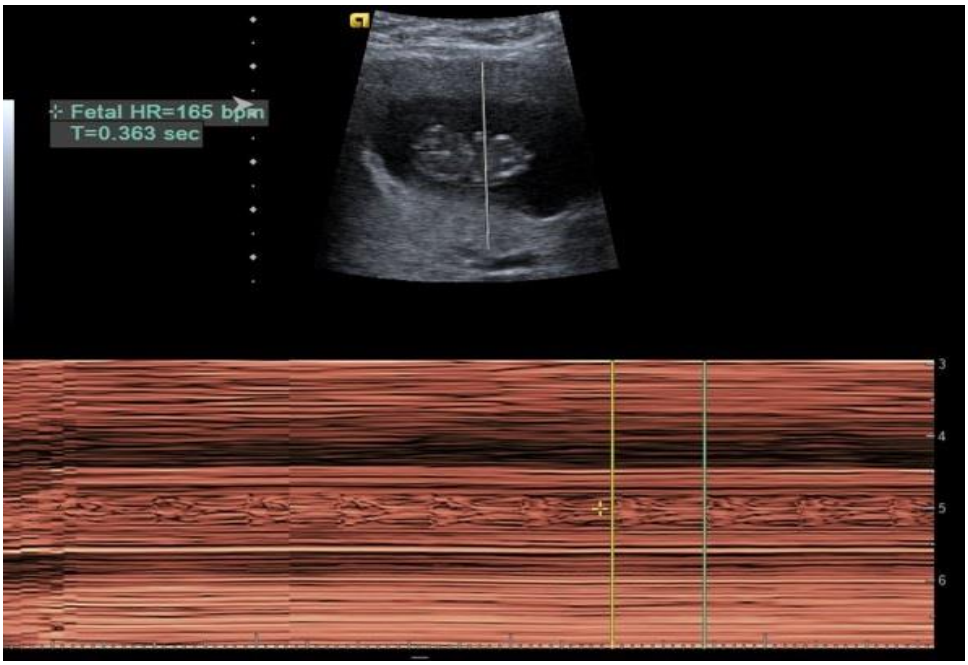
The fetal pole is measured as CRL.

CRL- Crown Rump Length.

Measurements have to be taken twice to confirm the baby's right size.



7. **Take a picture of the Fetal Heart (FH) beat in M-mode (NOT in power doppler, some facilities might ask you NOT to do a color picture of the heart)**



B) late Stage: 8-10 weeks gestation

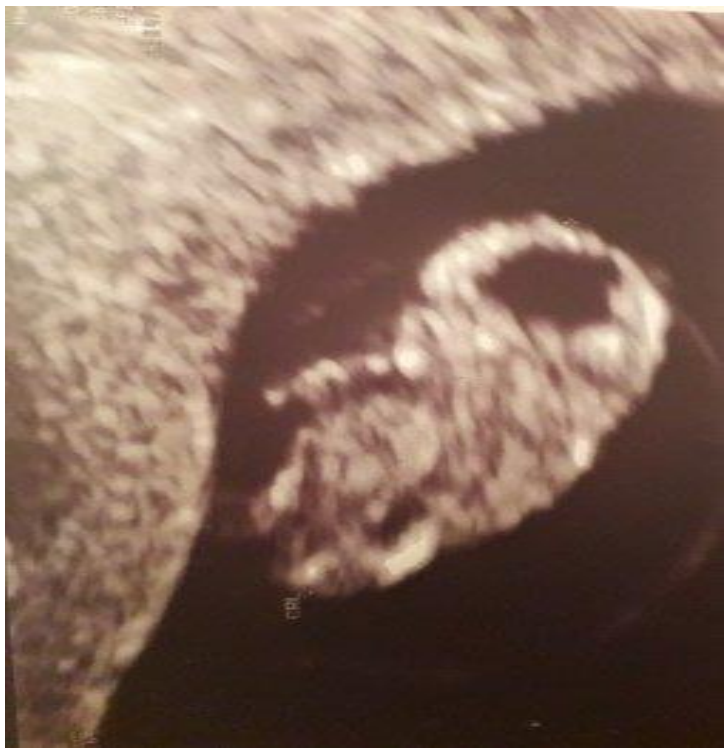
Doing the full protocol for the female pelvic ultrasound (Uterus/Ovaries) depends on the baby's size in this stage.

You will do all the above images and add: (this can be taken from an abdominal and/or transvaginal ultrasound approaches; it depends on the facility you work for).

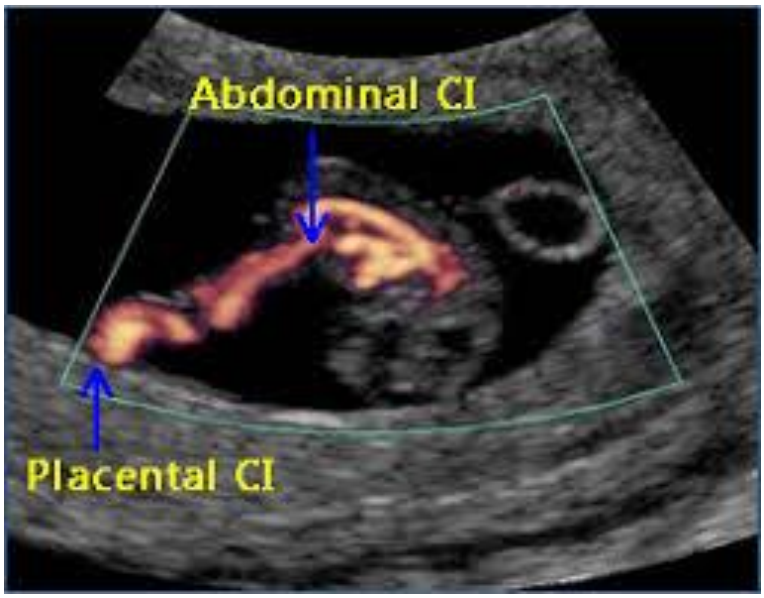
1. **Magnify the embryo' s head (NO Measurements are needed unless the Cisterna Magna (CM) is enlarged)**

   It is normal to see this cystic area in the baby's head at this stage, it is called the cisterna Magna and should change in appearance in the next trimesters.





2. **Cord Insertion (CI) to the embryo' s belly (one image is enough)**





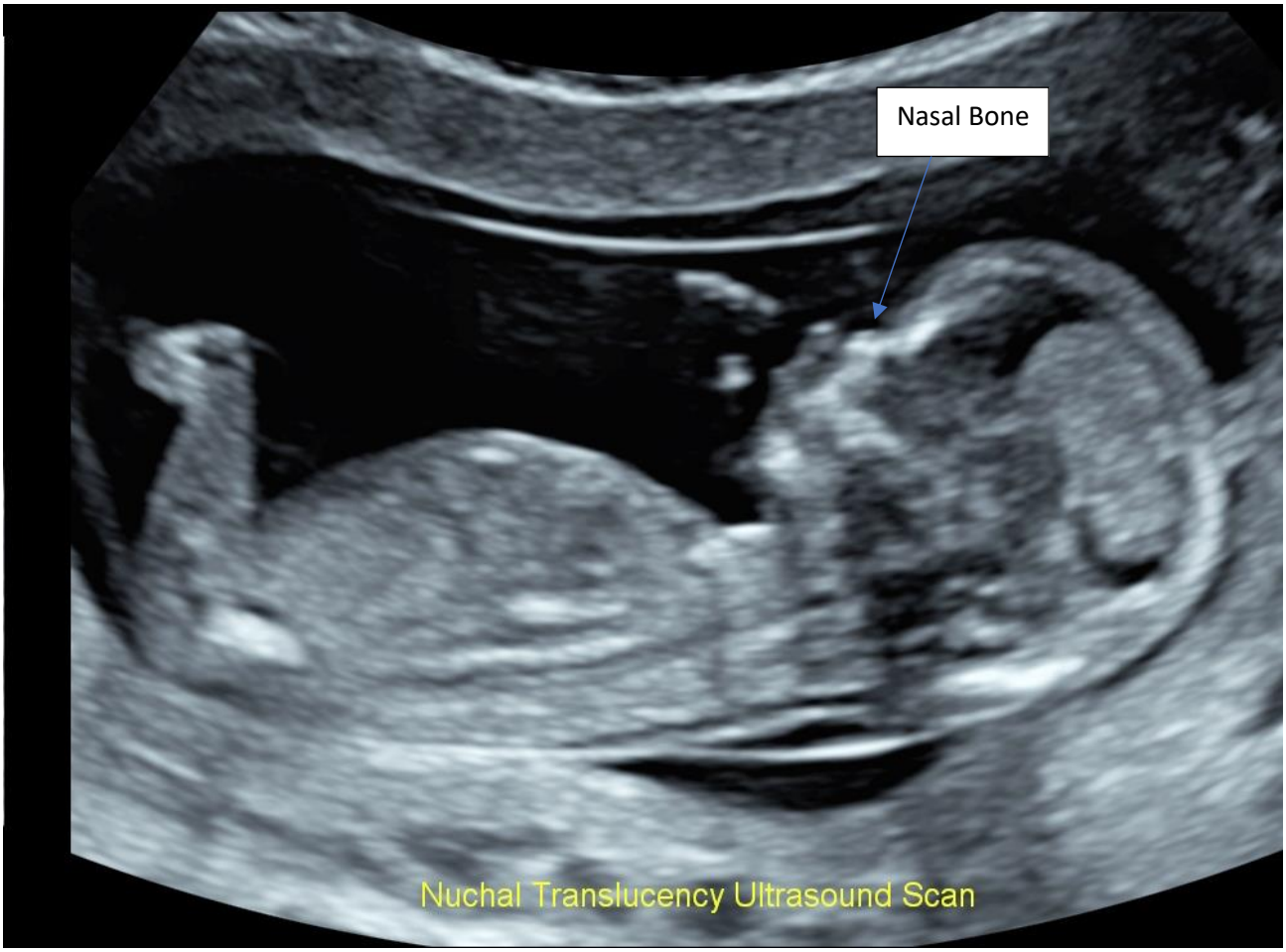


C) Latest Stage: 11-13 weeks and 6 days gestation

No need for a full protocol of the female pelvic ultrasound (Uterus/Ovaries), the embryo is big enough at this stage.

This stage is meanly for the Nuchal Translucency (transabdominal approach) and the Cervix Size (transvaginal approach), you can add measurements of the FP and the FH.

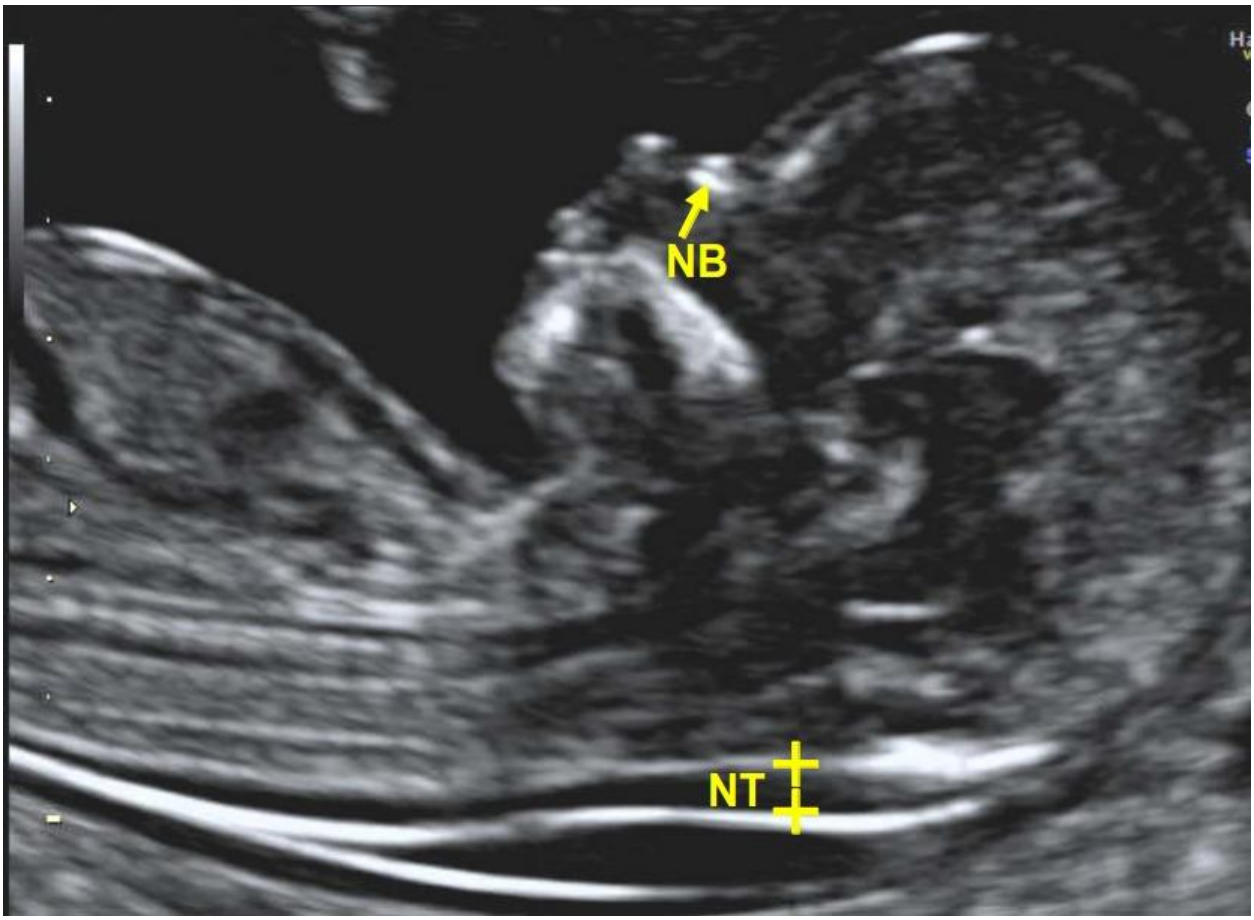
1. **Nuchal Translucency w/o Measurement-** The fetus should be in a neutral position, with the head in line with the spine and the nasal bone well seen.

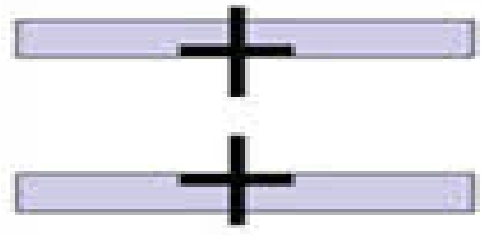


2. **Nuchal Translucency w/ Measurement.** **You will measure the nuchal fold thickness and is located at the back of the embryo' s neck.**

Note:

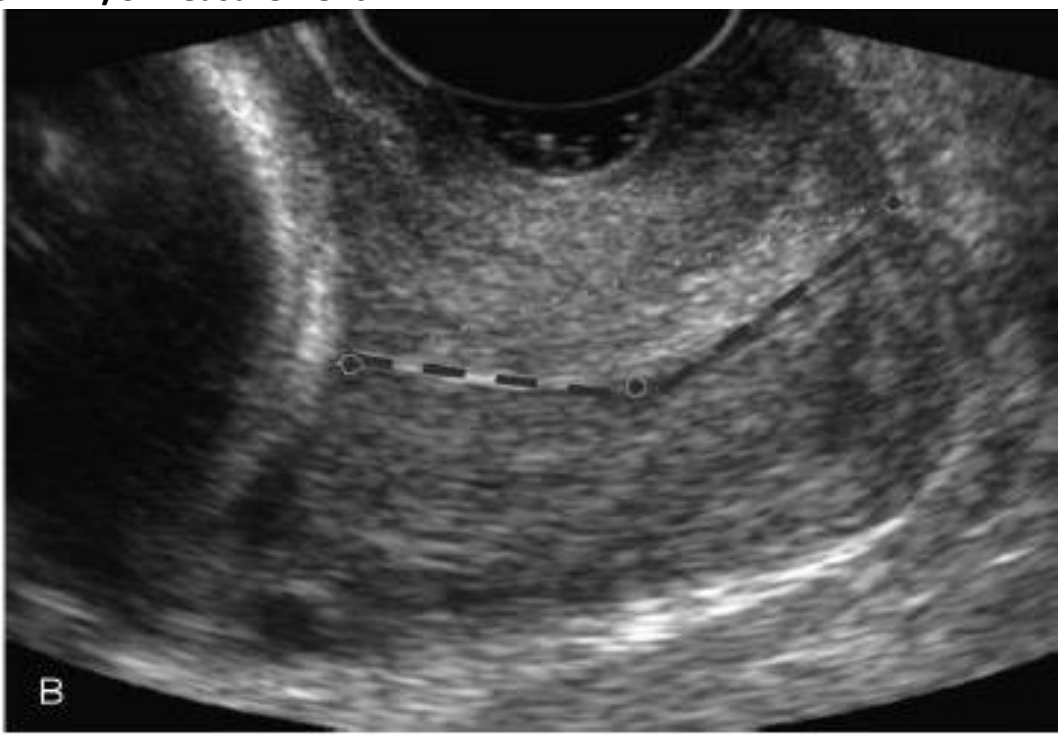
NT is measured from inner to inner.





Correct alignment

3. **Cervix w/o Measurement**



4. **Cervix w/ Measurement**



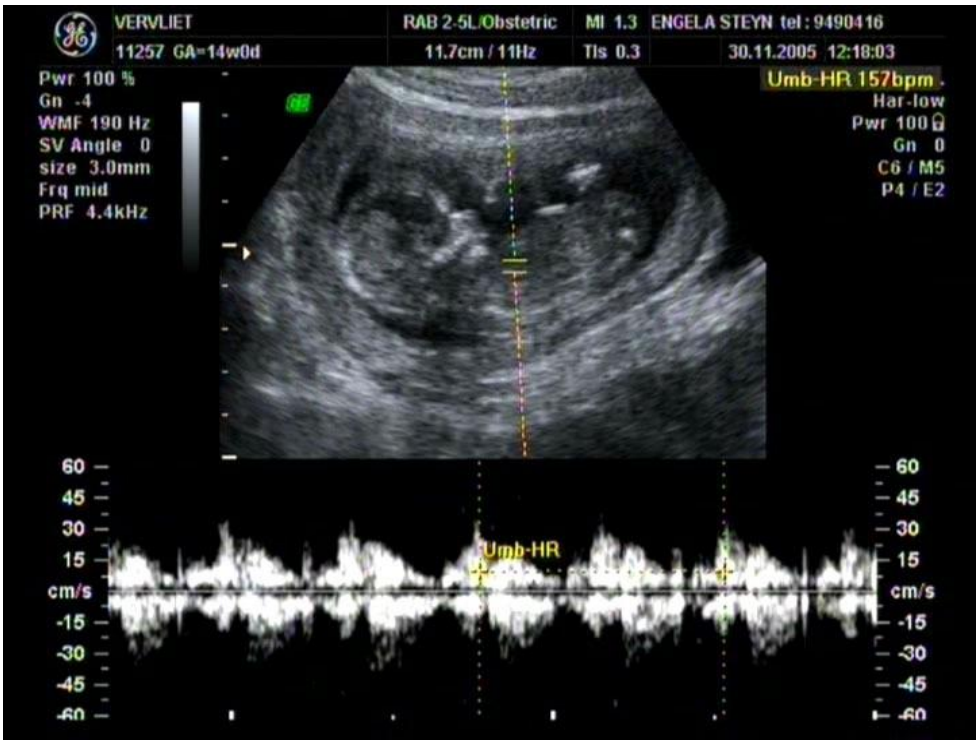
5. **Fetal Pole w/o Measurement**



6. **Fetal Pole w/ Measurement**



7. Take a picture of the Fetal Heart Beat. **It is ok to do a power doppler at this stage**



Remember every facility is different and the protocols might differ from place to another. This will be the basics of 1st trimester scanning.